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# CMCS 724 Final- Spring 2014

**Posted on 5/14/2014 5:30 pm Due by e-mail 48 hours later 5/16/2014 5:30 pm**

**1.** In the 35+ years of research in data modeling it became apparent that the models made "full circle". Identify the stages and the models introduced in each stage along with the needs they were trying to address each time. Were these needs addressed adequately? Or did they introduce new problems? Include in the transition for each of the following models:

Hierarchical → Network → Relational → Object-Relational Oriented → Semi-structured XML

**Ans**

Hierarchical model (IMS) was introduced first around 1968. It has the notation of record type which forces each instance to obey the data description. They are required to have a key. The record type must be arranged in a tree such that each record type has a unique parent record type. There are 2 problems. First, the information is repeated and may introduce inconsistency. Second, the existence of a record type depends on its parent which is not a case in real world that 'part' has to be associated with 'supplier'.

Network model (CODASYL) was introduced in 1969. Rather than organizing the record types as a tree, network model organizes the record types as a network. A given record can have multiple parents and an intermediate relation like 'supply' so that 'part' is okay to not associated with a 'supplier'. However, there are still drawbacks. First, the network is more complex than a tree so a programmer needs to navigate in a multi-dimensional hyperspace. Second, loading and recovering networks is more complex than hierarchies, for example about a ternary relationship, reasoning about 'supply' cannot be done without 'supplier' and 'part' which are the references.

Relational model was introduced in 1970 by Ted Codd at IBM. The motivation is to reduce time spent when programming with IMS. Relational model stores the data in a table which is a simple data structure. Access are done via high-level language. The table can represent almost anything including the n-ary relationship. Relational model utilizes the relational algebra so that it is simple for a high-level programmer to think and let the query optimizer does the work.

Object-Relational Oriented model was introduced for the need of GIS applications. It has some new functionalities that defining new data types, putting code in the database (stored procedure) and user-defined access methods are possible. However, this has not so much contributions for business data processing.

Semi-structured XML is the data that is not fully structured. The data is self-describing using tags and does not need a schema to be understood (schema last). However, XML is a schema language and the formatted documents are needed to be validated. XML can be hierarchical like in IMS, can have links like CODASYL, however, it is difficult to optimize things in XML such as building indexes and performing semantic matching in the case of heterogeneous data can be difficult.

**2.** While B-trees is the most successful access method for 1-d data it is not the case for higher-d data. a) Explain the reason for this. b) How does the R+tree structure attempts to solve it? c) Gist claims to be a one stop shop for (hierarchical) indexing. Can it simulate R+trees? And if so define the corresponding steps. Otherwise, argue it cannot be done.

**Ans**

a) B-trees are based on order so that the sorting can be done easily in one dimensional. For *n*-dimensional where *n*>1, the definition of order and sorting vanishes. So, an extension is required to fulfill this missing definition.
b) R+tree tries to approximate the region by its minimum bounding rectangle (MBR) and the MBR may break into several smaller MBRs and is stored in different leaf nodes.
c) Gist can simulate R+trees because Gist can simulate any height-balanced trees. Gist may logically use predicate to represent the bounding boxes. The search key has some methods as follows.
    - Consistent: the predicate of the subtree should not be overlapped with the query predicate.
    - Union: bounding box of all entries in a subtree.
    - Penalty: the cost is the size of the result-of-union tree minus the size of the tree.
    - pickSplit: use R+tree method.
    - Compress: form the MBR.
    - Decompress: identity function.

**3.** Query feedback has been proposed for Adaptive Buffer Management, Selectivity estimation, and Query Optimization. In all cases these techniques the underlying assumption is that each query is running in isolation. Can these methods be extended to account for multiple concurrent queries? What would be the requirements and/or hurdles for Multiple Query Buffer Management and Multiple Query Optimization?

**<u>Ans</u>**

Yes, they can. The usual requirements such as dead lock, access control or load balancing will present. The ACID properties of transactions should also be maintained. The scenario here is about multiple queries on one site which may be the same queries or different queries. Some common issues that may arise in concurrency control like lost update, dirty read and incorrect summary on buffer management should be considered as another requirement. Selectivity estimation should be extended as there will be more waiting time for another query to finish according to deadlock prevention.

**4.** Data Cube: a) What is the importance of the data cube? b) What is the number of all possible views captured in a cube if dimension columns have no levels within (flat dimensions), and c) estimate the number of views once again when the dimension columns d1, d2, …,dn have l1, l2, …, ln levels respectively? What does View Selection mean and why do we need to have it?

**<u>Ans</u>**

a) The data cube provides an efficient way to get aggregated statistics from multi-dimensional data which also generalizes group-by operator.
b) 2**$N$ where $N$ is the number of dimension columns.
c) (l1 + 1) * (l2 + 1) * …* (ln + 1).
d) View selection is to select the view to be materialized. It helps improve query performance to wisely choose a view to be materialized so that a query does not have to recompute the view every time it is executed.

**5.** Compare data pull and push concepts with those of querying and streaming data. b) What is the role of optimization in each of these paradigms? c) What is the role of indexing in these?

**<u>Ans</u>**

a) Pulling data is like querying data because these two actions share the definition of requesting the data from a server/database. Pushing data is like streaming data because these two actions share the definition of broadcasting the data from a server/data source.

b) Query/pulling optimization is to finding the best operator tree to execute the query. Pushing/streaming optimization is to finding the best broadcast program for a given underlying distribution, in other words, runtime query optimization.
c) For query/pulling, indexing is for an efficient disk scan. For pushing/streaming, indexing may be needed if the broadcasting program is not fixed to find the right track or chunk.

**6.** What are the similarity and the main difference between Aircache's temperature-based and LRU replacement strategy. Can Aircache use LRU instead? Explain your answer.

**Ans**

Aircache dynamically adapts its broadcast policy based on changes in item popularities (workloads). LRU is also a cache management techniques. However, LRU cannot replace Aircache. The reason is, in order to achieve adaptability, the information about 'air-cache hits' is not observable because it is not acknowledge by the clients. This makes LRU not applicable for the task. Aircache lies on 'air-cache misses' instead.

**7.** Explain the fundamentals and goals of the three approaches in data mining: (a) classification, clustering, and association. (b) Sketch a reduction of a classification scheme to discovering associations.

**Ans**

a) Classification is a supervised learning problem where the training data (measurements, observations, etc.) is given along with the labels which indicate the desired output for each data point. The classification problem tries to predict the category of the new data point based on the statistics in the training data.

   In contrast to classification, clustering is an unsupervised learning problem where the training data is given without labels. The task is to group the data point together based on some underlying relevant statistics which may reveal the existence of classes or clusters in the data.

   Association rule mining (market-basket analysis) is to find all the rules $X \rightarrow Y$ with minimum support and confidence threshold which represent interesting relation between the variable (or set of variables) $X$ and $Y$.
b) Decision tree is a good model to perform a reduction from association rule mining to classification. The resulting decision tree will have the variable X1 ^ X2 ^ … along the non-leaf nodes and the variable Y for the leaf node. (X1 ^ X2 ^ … -> Y) The support and confidence can be computer by node purity measures.

**8.** Explain the similarities and differences between data mining and machine learning.

**Ans**

Machine learning is a field that use some algorithms to automatically extract information or perform predictions based on training data. Most machine learning techniques are based on statistics. There are also some machine learning algorithms that are not based on statistics which may include searching.

Data mining focuses on discoveries of unknown properties in the data for knowledge discovery. The difference between machine learning and data mining is on the behavior of reproducing known knowledge and discovery unknown knowledge. In machine learning, supervised methods are likely to outperform unsupervised methods. While in data mining, labels are unlikely available so unsupervised learning methods are more likely to be deployed.

**9.** Assume that historical data is coming as a stream and tagged with timestamps before entered into an un-indexed database table. Each tuple of this table corresponds to an observed identifiable event (ala Aurora) and is repeated typically at known intervals, e.g. heart beat from a monitor identified by a patient, or the ticker of a stock. We are interested in a query that retrieves unusual behavior, namely if the heart beat decreased by more than 90%, or stock was trading at 30% off the previous value or at 300% the normal frequency. "Analyze" the requirements and "Sketch" an index method and its data structures that can quickly retrieve unusual behavior and the time it occurred. Note that the stream is a mix of hundreds or thousands of such events and there is no time to store and index the complete stream. Instead, we would like to index ONLY its unusual behavior.

**Ans**

The requirements should be that it can detect anomalies or some extreme trends in real time so that the emergency actions can be taken. The data may be at different level of temporal and cross sectional aggregation. We may keep the anomalies in 'store' and derive some information with 'scratch'. A sliding window can be used to track the recent context and build a 'sketch'. For indexing, I think just using a B-tree on the timestamp variable should be enough to retrieve the time where the anomalies occur.

For outlier or extreme trend detection, an R-tree whose MBR covers most of the data can be used so that it will be trivial to spot an outlier. An alternative may be multidimensional factor analysis (by using tensor decomposition) that can determine outliers/anomalies/novelties by measuring some statistics such as residual errors. The 'sketch' in this case may be some incremental statistics.

**10.** In your opinion, what are the most promising database research areas for advancing science? How will these enhance our knowledge and life in the next decade?

**Ans**

I think graph database is the most promising database research areas for advancing science. Since graph is so common and has a strong theory with a lot of sound applications, data mining with graph may reveal a lot of knowledge for many other fields such as social science, finance or biology. The graph connections will reveals new things without using much of the explicit and expensive join operation. Managing the data on another scale (BigData) will provide knowledge in the scale that a man or a group of men are not likely to do from the overwhelming computational power. The new knowledge will make people smarter, know what to do and what not to do, and improve the way we live. For example, we may know that a certain kind of food should not be eaten because it enhanced the risk of cancer and this will enhance our life span by some years.

**11.** Select one of the papers in the list below:

- May 6,8,13
- 16a.Crowed_Mining_2013crowd-1.pdf
- 16b.Value_InventionAG13.pdf
- 16c_MassiveTriangulation.pdf
- 16c.Massive.pdf
- 16d.Indexing_Moving_Objects_sigmod13.pdf
- 16e.Querying_Heterog_Inform_Using_Source_Descr_P251.pdf
- 16f.BreakingTheMemoryWall.pdf
- 16g.GPU_Terasort.pdf
- 16h.InformationExtraction_Sarawagi_Survey.pdf

And do the following: discuss the problem it addresses, describe the solution, evaluate the approach, and conclude whether the problem was addressed adequately. You can replace any of these papers with a newer or better that you have identified and feel it does a better job addressing the same problem.

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**Yael Amsterdamer and Yael Grossman and Tova Milo and Pierre Senellart,**
**"Crowd Mining", SIGMOD 2013.**

Sometimes we want to learn about new domains from the crowd but that data is not recorded anywhere such as folk medicine. Moreover, the contents in the new domain are unknown. In order to discover something interesting in the new domain, we need to ask some questions to the crowd to get the knowledge so "what to ask?" becomes a question.

Crowd data sourcing tries to collect the data from people by asking some questions to get the association rules. The model learns association rules of the form a,b -> c,d from each user as one personal database with transactions which are answers to the questions. The answer contains rule support, rule confidence and items.

Extensive questions are not feasible to be used to get the raw input for data mining. The reason is people do not remember comprehensive details, for example, "all cases treated by folk medicine in the past". So, the idea is we could instead ask a question about some short summaries for personally prominent patterns like "I treat patients with a cold every week" which will gives some aggregated statistics.

The authors propose the use of personal summaries to learn about general trends. A framework is proposed for effective implementation by using components and an experimental study as depicted in Figure 1. The components are black boxes and organized in a hierarchy so that each component will be associated with a step in the algorithm 1.

To get the summaries by asking questions, there are 2 types of questions to use, closed questions and open questions. Choosing an open or closed question (use [a]) is based on interpolating between obtaining new information and refining the accuracy of current estimations. This can be set by a predefined fixed ratio to find a good tradeoff between precision and recall. For closed question, the question is chosen based on grades (use [e]) of the interesting rules (enabled rules). To estimate the quality of the current output (use [f]) and the quality if asked about the new rule *r'*, a distribution is computed to reflect the probability for different values of the mean given the current knowledge (use [i]). Then, the enabled rule set is updated. After a desired number of iterations, a set of significant rules (use [j]) is outputted.

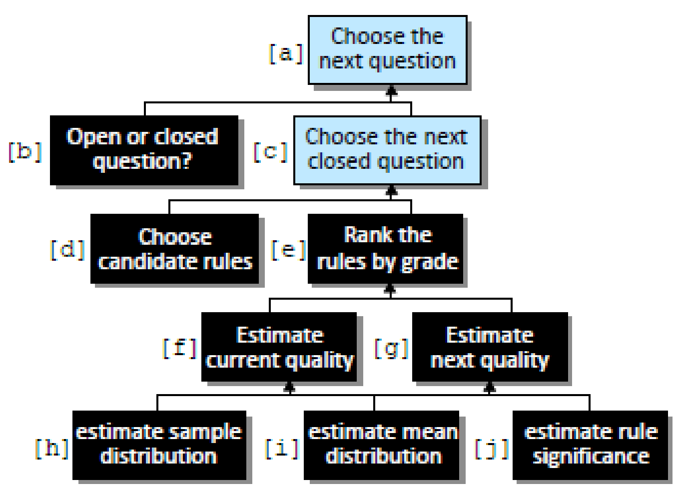


Figure 1: Framework Component Hierarchy

Therefore, the model of estimating the probability of making an error given the current knowledge is proposed. First, the distribution of the answer support and confidence is learned. Then, the significant by the position of a half of the distribution mass is estimated. Next, the error probability for the true mean is to be on the other side of the thresholds. Finally, the next question is picked to minimize the overall error.

The authors conduct the experiments on 3 datasets, synthetic, retail (market-basket analysis) and Wikipedia editing records. They compare their proposed crowd mining framework with two alternative baselines, random and greedy. They measure the results by using Precision, Recall and F-measure. They varies the parameters, such as the portion of open and closed questions, to observe the framework behavior.

There are 4 possible future works mentioned. First, *K* questions can be asked instead of just one question. Second, dependencies between rules when interpreting user answers should be taken into account so that more knowledge will be extracted and utilized from each answer. Third, the aggregation function can be another function rather than just average so that the experts should be weighted more. Fourth, the use of prior should be addressed on the data distribution or the users.

In my opinion, I think the problem was addressed in a very interesting way. However, user answers are in natural language so parsing the answers to get accurate meaning such that the association rules will really represent what the users want to tell is not addressed in this paper. Moreover, the trends and contradictions that may occur in the rule set, for example, a herb *H* have no side effects by an expert *A* versus a herb *H* will have some side effects by an expert *B*. Another example is an expert *A* may have an experience based on a time interval *t1* where an expert *B* may have his experience based on another time interval *t2* where the trends for each time interval are different. To give a specific example about the trends, people before and after the renaissance of neural networks, without updates, will have their opinion biased by the trend at a specific time they observed. These are possible because mining the data that is not record anywhere from people may lead to non-standardized answers. Even though the paper addresses error estimation of the rule, the distribution is assumed to be an unimodal Gaussian. The paper suggests to use a bimodal distribution but does not provide much details about this any further.